

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Circles

Circle questions are centre-radius questions first, then tangent or intersection questions.

CIRCLES · P2 · 03

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 CENTRE: Complete The Circle

TRIGGER *equation of a circle*

02 RADIUS: Distance To Centre

TRIGGER *point on circle, radius, diameter*

03 TANG: Tangent To Circle

TRIGGER *tangent at a point*

04 MEET: Line Meets Circle

TRIGGER *intersection of line and circle*

BANK EVIDENCE Local pages: 21 problem, 38 answer, 21 notes. Website primary entries: 19.

01 CENTRE: Complete The Circle

TRIGGER *equation of a circle*

BECOMES Complete the square in x and y .

FIRST LINE TO WRITE

$$(x - a)^2 + (y - b)^2 = r^2$$

SIMPLEST STRATEGY

- 1 Collect x and y terms.
- 2 Extract half coefficients.
- 3 New squares created.
- 4 Tidy constants.
- 5 Read centre.
- 6 Evaluate radius.

WORKED MODEL

$$x^2 + y^2 - 6x + 4y = 12 \Rightarrow (x - 3)^2 + (y + 2)^2 = 25.$$

HARD VARIANTS

- 1 If there is no $x^2 + y^2$ coefficient 1, divide first.
- 2 Radius squared must be positive; check impossible circle cases.
- 3 Use the completed form before drawing or finding tangents.

— BOTTOM LINE

The centre is read after completing both squares.

Practice 01

SAME IDEA Use CENTRE: Complete The Circle.

QUESTION

Find the centre and radius of $x^2 + y^2 + 8x - 2y = 8$.

YOUR SOLUTION

02 RADIUS: Distance To Centre

TRIGGER *point on circle, radius, diameter*

BECOMES Use distance from the centre.

FIRST LINE TO WRITE

$$r^2 = (x - a)^2 + (y - b)^2$$

SIMPLEST STRATEGY

- 1 Read the centre.
- 2 Apply distance formula.
- 3 Decide radius or diameter.
- 4 Insert point.
- 5 Use squared form if cleaner.
- 6 State exact radius.

WORKED MODEL

| Centre $(3, -2)$, point $(6, 2)$: $r^2 = 3^2 + 4^2 = 25$.

HARD VARIANTS

- 1 For diameter endpoints, the centre is the midpoint first.
- 2 Use r^2 in the circle equation, not r .
- 3 If a point is inside/outside, compare distance squared with radius squared.

— BOTTOM LINE

A point on a circle is a radius equation.

03 TANG: Tangent To Circle

TRIGGER *tangent at a point*

BECOMES Radius to tangent is perpendicular.

FIRST LINE TO WRITE

$$m_t = -\frac{1}{m_r}$$

SIMPLEST STRATEGY

- 1 Take centre and tangent point.
- 2 Assess radius gradient.
- 3 Negative reciprocal.
- 4 Generate line through the point.

WORKED MODEL

| If radius gradient is 2, tangent gradient is $-\frac{1}{2}$.

HARD VARIANTS

- 1 If the radius is vertical, the tangent is horizontal.
- 2 If the radius is horizontal, the tangent is vertical.
- 3 Check the tangent point lies on the circle if it is not stated.

— BOTTOM LINE

A circle tangent is a perpendicular line question.

04 MEET: Line Meets Circle

TRIGGER *intersection of line and circle*

BECOMES Substitute the line into the circle.

FIRST LINE TO WRITE

$$(x - a)^2 + (mx + c - b)^2 = r^2$$

SIMPLEST STRATEGY

- 1 Make y from the line.
- 2 Enter it into circle.
- 3 Evaluate the quadratic.
- 4 Turn roots into coordinates.

WORKED MODEL

| Substitute $y = x + 1$ into $(x - 2)^2 + y^2 = 10$.

HARD VARIANTS

- 1 Tangency means the quadratic has one repeated root.
- 2 No intersection means discriminant is negative.
- 3 Keep both roots unless the question restricts the point.

— BOTTOM LINE

Intersections need coordinates, not just x -values.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
equation of a circle	$(x - a)^2 + (y - b)^2 = r^2$
point on circle, radius, diameter	$r^2 = (x - a)^2 + (y - b)^2$
tangent at a point	$m_t = -\frac{1}{m_r}$
intersection of line and circle	$(x - a)^2 + (mx + c - b)^2 = r^2$